

BACHELOR'S THESIS PRESENTATION · UTCN · JULY 2026

Engineering an Ontology for Age-Related Macular Degeneration using Agentic AI and DL-Learner

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UTCN · AIRi Research Group · 2025–2026

Context and Motivation

What is AMD?

- ▶ Leading cause of blindness over 60
- ▶ 196M affected (2020), 288M by 2040
- ▶ Findings published daily on PubMed

The Knowledge Gap

- ▶ No updatable ontology for AMD
- ▶ MONDO / DOID: AMD as a single entry
- ▶ No agentic AI + DL-Learner combined

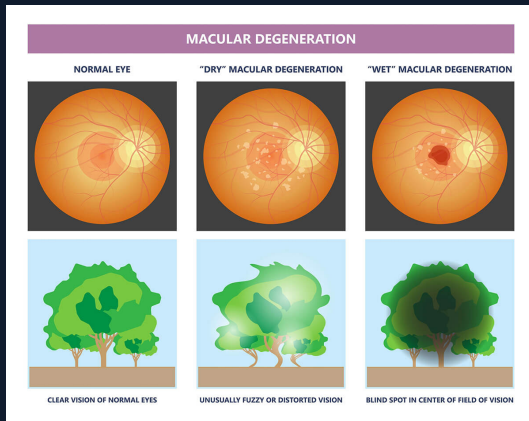


Figure 5.1: Retinal changes in AMD (Bright Sight Eye Specialists)

Challenges

Ontology Engineering

- ▶ Empty classes, no triples (SCHEMA-MINERpro)
- ▶ Models invent names freely (SCHEMA-MINERpro)
- ▶ Generated ontologies need validation (NeOn-GPT)
- ▶ Context window overflows (SCHEMA-MINERpro)

Agentic AI & DL-Learner

- ▶ Agents must avoid conflicting updates (ReAct)
- ▶ Agentic retrieval needs confidence (AMG-RAG)
- ▶ DL-Learner needs a sound ontology
- ▶ No system combines both approaches

Proposed Solution

Three LangChain agents · OWL 2 · HermiT · DL-Learner

Proposed Solution: System Architecture



Proposed Solution: Discovery & Extraction

Agent 1: Literature Discovery

- ▶ Autonomous **PubMed** search
- ▶ Relevance gate before extraction
- ▶ Only relevant abstracts advance

Agent 2: Knowledge Extraction

- ▶ One abstract at a time, tool calls
- ▶ Guided by a **compact summary**
- ▶ Stages: baseline $\rightarrow$ 20 $\rightarrow$ 100 abstracts

Structural Constraints

Failure	Design choice
Context overflow	Compact summary
Spelling duplicates	Fuzzy dedup (≤ 2)
Class explosion	Pending queue (≥ 2)
Predicate explosion	Closed vocabulary (9)
Hallucinations	Hallucination guard

Proposed Solution: Validation & Human Review

Agent 3: Validation

- ▶ Global checks across the ontology
- ▶ Cross-abstract duplicate detection
- ▶ Domain / range & disjointness checks
- ▶ Proposes fixes, never auto-commits

Human-in-the-Loop Interface

- ▶ **FastAPI** backend + **Vue 3** frontend
- ▶ Real-time agent output streaming
- ▶ Expert **approves** / **rejects** each change

Key design principle

No change committed without expert approval

Results: Extracted Ontology

Final Ontology

- ▶ 11 classes, 104 instances, 114 triples
- ▶ From 100 AMD abstracts

Class	Instances
Treatment + AntiVEGF	31
Biomarker + Gene	24
DiagnosticMethod	15
MolecularTarget	11
ClinicalOutcome	9
Disease + subtypes	7
RiskFactor	7

HermiT Reasoning

- ▶ Full logical consistency confirmed
- ▶ 38 axioms inferred by HermiT

DL-Learner: Selected Discoveries

Expression	Acc
AntiVEGFTherapy $\sqcap$ treats $\exists$ Disease $\sqcap$ inhibits $\exists$ MolTarget	100%
Biomarker $\sqcap$ associatedWith $\exists$ Disease	100%

Structural Results

HermiT Reasoner

- ✓ Full consistency, 0 unsatisfiable
- ✓ 38 axioms inferred

OOPS! Pitfall Scanner

- ✓ Only 3 minor pitfalls, none critical

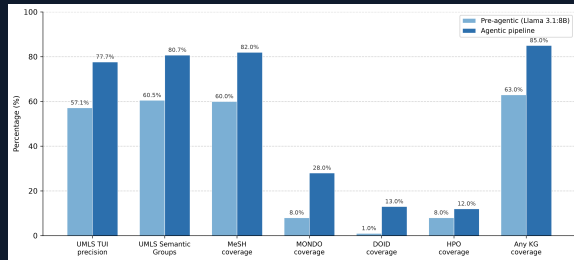
Themis Test Suite

Test category	Passed
Class declarations	7/7
Property declarations	9/9
Domain constraints	9/9
Range constraints	9/9
Disjointness	3/3
Subclass relationships	3/3
Individual classifications	10/10
Total	50/50

Semantic Results

UMLS & External KGs

- ▶ +20pp UMLS precision vs. baseline
- ▶ 85% of entities in a KG (was 63%)
- ▶ Gains: MONDO 8→28%, DOID 1→13%
- ▶ Remaining 15%: trade names



Pre-agentic baseline vs. agentic pipeline

Competency Questions

- ▶ 28% of 143 questions answered
- ▶ Strong on treatment, weak on lifestyle

DL-Learner Results

27 Experiments, 4 Categories

- ▶ 27 run, 17 reported in detail
- ▶ Avg F1/F2/F3 $\approx$ 92% (flat = robust)
- ▶ 6/17 problems at 100%
- ▶ CELOE stable; OCEL crashed

One Informative Failure

- ▶ imaging_vs_functional: 50%
- ▶ Missing triples = detected gap

Selected Learned Expressions

Expression	Acc
AntiVEGFTherapy $\sqcap$ treats $\exists$ Disease $\sqcap$ inhibits $\exists$ MolTarget	100%
Biomarker $\sqcap$ associatedWith $\exists$ Disease	100%
AntiVEGFTherapy $\sqcup$ (treats $\geq$ 3 Thing)	94.7%

Recovered from structure alone

Clinical Expert Validation

Confirmed by Ophthalmologist

- ✓ 7-class structure fits AMD
- ✓ 3 subtypes + OCT correct
- ✓ Anti-VEGF drugs correct
- ✓ Age & smoking as risk factors

Refinements Suggested

- ▶ Blindness not for DryAMD
- ▶ Uveitis link still debated
- ▶ Drusen also a clinical outcome
- ▶ Add newer agents (Faricimab)

Medically grounded and structurally sound

Conclusions

All Objectives Achieved

- O_1 ✓ Autonomous AMD ontology updating
- O_2 ✓ SCHEMA-MINERpro baseline evaluated
- O_3 ✓ Three LangChain agents implemented
- O_4 ✓ HITL web interface deployed
- O_5 ✓ OWL 2 + HermiT: full consistency
- O_6 ✓ DL-Learner: 91.3% avg accuracy
- O_7 ✓ UMLS + clinical validation

Personal Contributions

What I built and delivered

- ▶ Three-agent LangChain pipeline from scratch
- ▶ Human-in-the-loop interface (FastAPI + Vue 3)
- ▶ OWL 2 export + HermiT integration
- ▶ UMLS & external KG evaluation
- ▶ Clinical validation by an ophthalmologist
- ▶ 27 DL-Learner experiments, first for AMD
- ▶ Ontology + FAIR dataset published
- ▶ Paper to RuleML+RR 2026 Rule Challenge

Thank You!

Demo: calinaborzan-amd-ontology.hf.space
Ontology: w3id.org/amd-ontology
Code: github.com/CalinaBorzan/AMD-ontology
Dataset: doi.org/10.6084/m9.figshare.32301858
Paper: RuleML+RR 2026 Rule Challenge

Limitations

- ! Continuous update is proof-of-concept (fixed dataset, not live)
- ! Coverage limited by corpus: 28% of patient questions
- ! HITL review is a bottleneck at scale
- ! Higher compute cost: ~ 45 min/run vs. single-model

Future Work

Immediate Extensions

- ▶ Extend dataset to full-text articles
- ▶ Confidence scoring to reduce review
- ▶ Add imaging-to-biomarker triples

Longer-Term Directions

- ▶ Migrate storage to GraphDB
- ▶ Add a vector database layer
- ▶ Extend to diabetic retinopathy, glaucoma

Selected Bibliography

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